

Definitions in `Sarason/` and `R2.lean`

Lean Formalization Project

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This document provides a comprehensive list of all definitions manually formalized in the `ComplexAnalysis` project, specifically within `R2.lean` and the `Sarason/` directory. No axioms are used; all concepts are built strictly from $\mathbb{R}^2$ foundations.

1 `ComplexAnalysis/R2.lean`

These definitions establish the pure $\mathbb{R}^2$ foundational layer, strictly avoiding Mathlib topological/metric dependencies where requested, favoring $\epsilon - \delta$ constructions.

- `sqNorm` ($x \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The squared Euclidean norm.

$$\text{sqNorm}(x) = x_1^2 + x_2^2$$

- `sqDist` ($x, y \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The squared Euclidean distance.

$$\text{sqDist}(x, y) = \text{sqNorm}(x - y)$$

- `euclideanNorm` ($x \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The standard Euclidean norm.

$$\text{euclideanNorm}(x) = \sqrt{x_1^2 + x_2^2}$$

- `euclideanDist` ($x, y \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The Euclidean distance.

$$\text{euclideanDist}(x, y) = \sqrt{\text{sqDist}(x, y)}$$

- `LimitRtoR2` ($f : \mathbb{R} \rightarrow \mathbb{R}^2, a \in \mathbb{R}, L \in \mathbb{R}^2$):

$$\forall \epsilon > 0, \exists \delta > 0, \forall x, 0 < |x - a| < \delta \implies \text{euclideanDist}(f(x), L) < \epsilon$$

- `LimitR2toR` ($f : \mathbb{R}^2 \rightarrow \mathbb{R}, a \in \mathbb{R}^2, L \in \mathbb{R}$):

$$\forall \epsilon > 0, \exists \delta > 0, \forall x, 0 < \text{euclideanDist}(x, a) < \delta \implies |f(x) - L| < \epsilon$$

- `LimitR2toR2` ($f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, a \in \mathbb{R}^2, L \in \mathbb{R}^2$):

$$\forall \epsilon > 0, \exists \delta > 0, \forall x, 0 < \text{euclideanDist}(x, a) < \delta \implies \text{euclideanDist}(f(x), L) < \epsilon$$

- **ConvergesR2** ($\text{seq} : \mathbb{N} \rightarrow \mathbb{R}^2, L \in \mathbb{R}^2$):

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, \text{euclideanDist}(\text{seq}(n), L) < \epsilon$$

- **HasFDerivAt_R2_eps** ($u, v : \mathbb{R}^2 \rightarrow \mathbb{R}, u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}, a \in \mathbb{R}^2$): Defines the Fréchet derivative of $F(x, y) = (u(x, y), v(x, y))$ at a . Let $L(h) = \begin{pmatrix} u_{x0} & u_{y0} \\ v_{x0} & v_{y0} \end{pmatrix} h$.

$$\lim_{h \rightarrow 0} \frac{\text{euclideanNorm}(F(a + h) - F(a) - L(h))}{\text{euclideanNorm}(h)} = 0$$

- **DifferentiableAt_R2_eps** ($u, v : \mathbb{R}^2 \rightarrow \mathbb{R}, a \in \mathbb{R}^2$):

$$\exists u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}, \text{HasFDerivAt_R2_eps}(u, v, u_{x0}, u_{y0}, v_{x0}, v_{y0}, a)$$

- **HasDerivAt_RtoR2_eps** ($f : \mathbb{R} \rightarrow \mathbb{R}^2, f'_0 \in \mathbb{R}^2, t_0 \in \mathbb{R}$): Defines the derivative of a parametric curve in $\mathbb{R}^2$.

2 ComplexAnalysis/Sarason/Definitions.lean

This file bridges the $\mathbb{R}^2$ real-differentiability into $\mathbb{C}$ -differentiability using epsilon-delta constructions.

- **HasLimitAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, L \in \mathbb{C}, z_0 \in \mathbb{C}$):

$$\forall \epsilon > 0, \exists \delta > 0, \forall z, 0 < |z - z_0| < \delta \implies |f(z) - L| < \epsilon$$

- **HasLimitAtInf_eps** ($f : \mathbb{C} \rightarrow \mathbb{R}, L \in \mathbb{R}$):

$$\forall \epsilon > 0, \exists R > 0, \forall z, |z| > R \implies |f(z) - L| < \epsilon$$

- **HasDerivAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, f' \in \mathbb{C}, z_0 \in \mathbb{C}$): Complex differentiability at a point.

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'$$

- **DifferentiableAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}$):

$$\exists f' \in \mathbb{C}, \text{HasDerivAt_eps}(f, f', z_0)$$

- **HasPartialDerivX_C_to_R_eps** / **HasPartialDerivY_C_to_R_eps**: Partial derivatives of a real-valued function of a complex variable (identifying $\mathbb{C} \cong \mathbb{R}^2$).

- **deriv_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}, h : \text{DifferentiableAt_eps } f \ z_0$): The classical derivative value $f'(z_0)$ extracted via choice.

- **HasDerivAt_R_to_C_eps** ($f : \mathbb{R} \rightarrow \mathbb{C}, f'_0 \in \mathbb{C}, t_0 \in \mathbb{R}$): Derivative of a complex-valued path parameterized by a real variable.

- **HolomorphicAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}$): f is complex-differentiable in some open neighborhood around z_0 .

- **HolomorphicOn_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, G \subset \mathbb{C}$): f is holomorphic at every point in the open set G .

3 ComplexAnalysis/Sarason/Chapter1.lean

These cover the basic topology of the extended complex plane and stereographic projections.

- **chordalMetric** ($z, w \in \mathbb{C} \rightarrow \mathbb{R}$): The chordal distance between two points in the complex plane mapped to the Riemann sphere.

$$\chi(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}$$

- **chordalMetricToInf** ($z \in \mathbb{C} \rightarrow \mathbb{R}$): The chordal distance from a point z to the point at infinity ∞ .

$$\chi(z, \infty) = \frac{2}{\sqrt{1 + |z|^2}}$$

- **Φ** ($z \in \mathbb{C} \rightarrow \mathbb{R}^3$): The stereographic projection mapping z to a point on the unit sphere $S^2 \subset \mathbb{R}^3$.
- **ChordalC**: Type alias for $\mathbb{C}$ equipped with the chordal metric.

4 ComplexAnalysis/Sarason/Chapter2.lean

These house the core objects for calculus, complex derivatives (Wirtinger derivatives), paths, and conformality.

- **del / delBar** ($f : \mathbb{C} \rightarrow \mathbb{C}, z \in \mathbb{C}$): Wirtinger derivatives.

$$\partial f = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right), \quad \bar{\partial} f = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right)$$

- **del_eps / delBar_eps** ($u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}$): Wirtinger derivatives expressed purely in terms of the real partial derivatives of $f = u + iv$.

$$\partial f = \frac{1}{2}(u_{x0} + v_{y0}) + \frac{i}{2}(v_{x0} - u_{y0}), \quad \bar{\partial} f = \frac{1}{2}(u_{x0} - v_{y0}) + \frac{i}{2}(v_{x0} + u_{y0})$$

- **path_in_C** ($x, y \in \mathbb{C}$): A continuous path $\gamma : [0, 1] \rightarrow \mathbb{C}$ starting at x and ending at y .
- **translated_path** ($c, x, y \in \mathbb{C}, \gamma : \text{path_in_C } x \ y$): A path shifted by the constant c , creating a path from $c + x$ to $c + y$.
- **pathDeriv** ($\gamma : \text{path_in_C } x \ y, t \in \mathbb{R}$): The tangent vector $\gamma'(t) \in \mathbb{C}$, extracted via choice from **HasDerivAt_R_to_C_eps** extended to $\mathbb{R}$.
- **pathDirection** ($\gamma : \text{path_in_C } x \ y, t \in \mathbb{R}$): The argument (angle) of the tangent vector, $\arg(\gamma'(t))$.
- **pathAngle** ($\gamma_1, t_1, \gamma_2, t_2$): The angle between two intersecting paths.

$$\text{pathAngle}(\gamma_1, t_1, \gamma_2, t_2) = \text{pathDirection}(\gamma_2, t_2) - \text{pathDirection}(\gamma_1, t_1)$$

- **conformal** ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}$): f preserves angles between any two paths intersecting at z_0 with non-zero derivatives.
- **centered_line_path** ($z, v \in \mathbb{C}$): A straight-line path from $z - v$ to $z + v$ evaluating to z at $t = 1/2$.

$$\gamma(t) = z + (2t - 1)v$$

5 ComplexAnalysis/Sarason/Chapter3.lean

Matrix representations of Möbius transformations acting on the Riemann sphere.

- `translationMatrix` ($b \in \mathbb{C}$):

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

- `scalingMatrix` ($a \in \mathbb{C}, a \neq 0$):

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

- `inversionMatrix`:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$